

Analysis of Urban Growth Dynamics in the United Arab Emirates

Using Multi-Temporal Sentinel-2 Satellite Imagery
(2018–2024)



Capstone Project Report

Land Cover Classification and Change Detection Workflow

Study Period:

2018 – 2024

Data Source:

Sentinel-2 MSI

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Abstract

This report presents a comprehensive methodology for analyzing urban growth dynamics across the United Arab Emirates (UAE) using multi-temporal Sentinel-2 satellite imagery spanning the period 2018–2024. The study employs a supervised machine learning approach based on the Random Forest algorithm to classify land cover into four primary categories: urban/built-up areas, vegetation, desert, and water bodies.

The workflow integrates Google Earth Engine (GEE) for large-scale satellite data processing and composite generation, OpenStreetMap (OSM) vector data for training label extraction, and Kaggle notebooks for machine learning model development and prediction. The UAE study area was strategically divided into ten spatial tiles to overcome computational limitations and enable efficient processing of high-resolution multi-spectral data.

Key spectral indices including NDVI, NDBI, MNDWI, BSI, and GHSL were derived alongside raw Sentinel-2 bands to enhance discriminative capacity between land cover classes. The resulting classified rasters provide a quantitative basis for assessing spatial patterns of urban expansion, vegetation change, desert dynamics, and hydrological changes over the six-year study period. This methodology demonstrates the potential of combining cloud-based geospatial processing with open-source training data for operational land cover monitoring in arid environments.

Keywords: Urban Growth, Land Cover Classification, Sentinel-2, Random Forest, UAE, Remote Sensing, Change Detection, Google Earth Engine, OpenStreetMap

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1 Introduction

1.1 Background and Motivation

The United Arab Emirates has experienced one of the most rapid urbanization rates globally over the past few decades. The transformation from desert landscapes to modern metropolitan areas, particularly in Dubai and Abu Dhabi, represents a significant case study in urban growth dynamics. Understanding the spatial and temporal patterns of this expansion is crucial for sustainable urban planning, environmental management, and resource allocation.

Remote sensing technology, particularly satellite imagery, offers an unparalleled capability to monitor land cover changes over large geographic areas with consistent temporal coverage. The European Space Agency's Sentinel-2 mission provides free, high-resolution multi-spectral imagery that is particularly well-suited for land cover classification and change detection studies.

1.2 Study Objectives

This capstone project aims to:

1. **Develop an automated workflow** for land cover classification using Sentinel-2 imagery and machine learning techniques.
2. **Map urban, vegetation, desert, and water classes** across the entire UAE for the years 2018, 2021, and 2024.
3. **Quantify urban growth patterns** by comparing classified outputs across the three time periods (2018–2021, 2021–2024, 2018–2024).
4. **Demonstrate the integration** of cloud-based processing (GEE), open-source labels (OSM), and machine learning (Random Forest) for operational land cover monitoring.

1.3 Study Area

The United Arab Emirates is located on the eastern part of the Arabian Peninsula, covering approximately 83,600 km². The terrain is predominantly desert, with coastal plains along the Persian Gulf and the Gulf of Oman. The country comprises seven emirates, with the most significant urban concentrations in:

- **Dubai** – Major commercial and tourism hub
- **Abu Dhabi** – Capital and largest emirate by area
- **Sharjah** – Cultural and industrial center
- **Ras Al Khaimah, Fujairah, Ajman, Umm Al Quwain** – Smaller urban centers

The arid climate, characterized by extremely low precipitation (<150 mm annually) and high temperatures, presents unique challenges for land cover classification, particularly in distinguishing between desert and built-up areas using spectral signatures alone.

1.4 Report Structure

This report is organized as follows: Section 2 details the complete methodological workflow; Section 3 describes the data sources and preprocessing steps; Section 4 explains the machine learning implementation; Section 5 presents the classification results and change detection analysis; and Section 7 provides conclusions and recommendations for future work.

2 Methodology

2.1 Workflow Overview

The overall workflow consists of six major stages, as illustrated in Figure 1. Each stage builds upon the previous one, creating a pipeline from raw satellite data to final classified land cover maps.

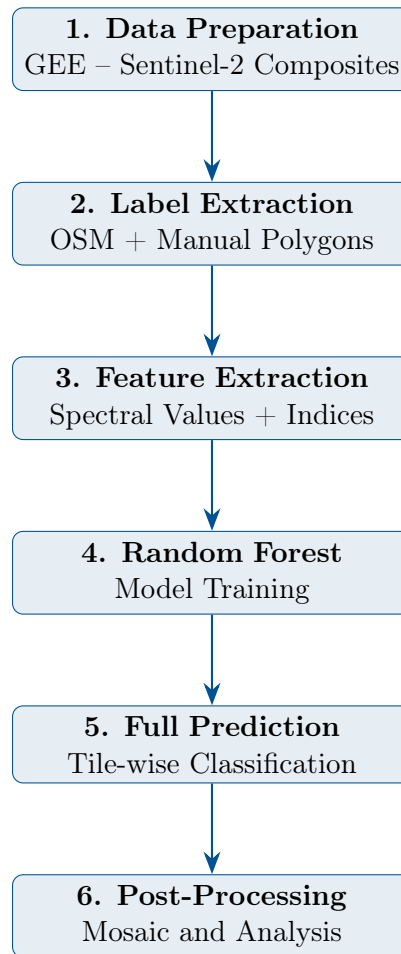


Figure 1: UAE Urban Growth Classification Workflow – Six-stage pipeline from satellite data to classified land cover maps.

2.2 Stage 1: Data Preparation in Google Earth Engine

The first stage involves processing Sentinel-2 imagery within the Google Earth Engine (GEE) platform. GEE was selected for its cloud-computing capabilities, which enable processing of large spatial extents without local hardware limitations.

2.2.1 Spatial Tiling Strategy

To manage computational constraints and memory limitations inherent in processing the entire UAE at once, the study area was divided into **10 spatial tiles**. This tiling strategy offers several advantages:

- Reduces per-batch memory requirements
- Enables parallel processing across tiles
- Facilitates error isolation and quality control
- Allows for incremental workflow development and testing

2.2.2 Sentinel-2 Composite Generation

For each tile, a cloud-free composite image was generated using the Sentinel-2 MultiSpectral Instrument (MSI) Level-2A surface reflectance product. The compositing period was defined as the full calendar year (2018, 2021, and 2024 respectively), with a cloud cover threshold of $<20\%$ applied to individual scenes.

Each exported composite contains the following **10 spectral bands and derived features**:

Table 1: Spectral Bands and Derived Indices Used for Classification

No.	Feature	Description
1	B2 (Blue)	490 nm – Atmospheric correction and water body detection
2	B3 (Green)	560 nm – Vegetation health assessment
3	B4 (Red)	665 nm – Chlorophyll absorption, vegetation stress
4	B8 (NIR)	842 nm – Vegetation vigor, biomass estimation
5	B11 (SWIR)	1610 nm – Soil moisture, built-up area detection
6	NDVI	Normalized Difference Vegetation Index – Vegetation density
7	NDBI	Normalized Difference Built-up Index – Built-up area detection
8	MNDWI	Modified Normalized Difference Water Index – Water body enhancement
9	BSI	Bare Soil Index – Soil and desert detection
10	GHSL	Global Human Settlement Layer – Settlement presence proxy

2.2.3 Index Formulations

The derived spectral indices were calculated using the following standard formulations:

$$\text{NDVI} = \frac{\text{B8} - \text{B4}}{\text{B8} + \text{B4}} \quad (1)$$

$$\text{NDBI} = \frac{\text{B11} - \text{B8}}{\text{B11} + \text{B8}} \quad (2)$$

$$\text{MNDWI} = \frac{\text{B3} - \text{B11}}{\text{B3} + \text{B11}} \quad (3)$$

$$\text{BSI} = \frac{(\text{B11} + \text{B4}) - (\text{B8} + \text{B2})}{(\text{B11} + \text{B4}) + (\text{B8} + \text{B2})} \quad (4)$$

2.3 Stage 2: Training Label Extraction

Training labels for supervised classification were derived from a combination of OpenStreetMap (OSM) vector data and manually digitized polygons in QGIS.

2.3.1 Urban Class (Class 1)

Urban and built-up areas were identified using OSM road network data and related infrastructure polygons. The following `fclass` values were included:

Listing 1: OSM Query for Urban Class

```

1  "fclass" IN (
2      'motorway', 'motorway_link',
3      'trunk', 'trunk_link',
4      'primary', 'primary_link',
5      'secondary', 'secondary_link',
6      'tertiary', 'tertiary_link',
7      'residential', 'service',
8      'living_street', 'busway',
9      'unclassified'
10 )

```

These features were assigned to **Class 1 = Urban/Built-up**. The rationale for using road networks as proxies for urban areas is based on the high spatial correlation between road density and built-up land cover in urban environments.

2.3.2 Vegetation Class (Class 2)

Vegetation areas were identified using OSM landuse and natural feature classifications:

Listing 2: OSM Query for Vegetation Class

```

1  "fclass" IN (
2      'orchard', 'forest', 'farmland',
3      'park', 'meadow', 'allotments',
4      'grass', 'recreation_ground',
5      'farmyard'
6  )
7  -- OR --
8  "fclass" = 'tree'

```

These features were assigned to **Class 2 = Vegetation**.

2.3.3 Desert Class (Class 3)

Desert polygons were manually digitized in QGIS across the major desert landscapes of the UAE, including sand dune fields, gravel plains, and salt flats, ensuring spectral diversity within the class. These vector layers were rasterized and incorporated into the training dataset to improve classification accuracy for this dominant land cover type.

2.3.4 Water Bodies Class (Class 4)

Water body polygons were manually delineated in QGIS to supplement OSM-derived water features, capturing reservoirs, artificial lakes, and coastal water bodies that may have been incomplete or outdated in the OSM database. These vector layers were rasterized and merged with the OSM-derived labels to improve training data quality for the water class.

2.4 Stage 3: Feature Extraction in Kaggle

The machine learning workflow was implemented in a Kaggle notebook environment, providing access to computational resources for model training and prediction.

2.4.1 Rasterization of Labels

OSM vector labels and manually digitized polygons were rasterized to create label rasters spatially aligned with the Sentinel-2 composite imagery. Each pixel in the label raster received one of the following class values:

Table 2: Land Cover Class Encoding

Class Value	Land Cover	Description
1	Urban/Built-up	Buildings, infrastructure
2	Vegetation	Parks, farms, forests, grasslands
3	Desert	Sand dunes, gravel plains, salt flats
4	Water	Lakes, reservoirs, coastal waters

2.4.2 Training Dataset Construction

For every labeled pixel, the corresponding Sentinel-2 spectral and index values were extracted, creating a structured training dataset where:

- **Input features (X):** B2, B3, B4, B8, B11, NDVI, NDBI, MNDWI, BSI, GHSL
- **Target variable (y):** Land cover class (1, 2, 3, or 4)

2.4.3 Label Rasterization Summary

The rasterization process produced the following pixel distribution across the four land cover classes:

Table 3: Training Label Rasterization Summary

Metric	Value
Total pixels	110,166,016
Labeled pixels	1,687,662

Table 4: Class Distribution in Training Labels

Class	Pixel Count	Percentage
Class 1 – Urban	1,080,777	64.0%
Class 2 – Vegetation	14,689	0.9%
Class 3 – Desert	499,543	29.6%
Class 4 – Water	92,653	5.5%
Total	1,687,662	100.0%

To address class imbalance and ensure robust model training, a stratified sampling approach was applied. The final training dataset comprised:

Table 5: Final Training Sample Selection

Class	Samples Selected	Available
Class 1 – Urban	200,000	1,080,777
Class 2 – Vegetation	14,689	14,689
Class 3 – Desert	200,000	499,543
Class 4 – Water	92,653	92,653
Total	507,342	–

2.5 Stage 4: Random Forest Classification

A Random Forest (RF) classifier was trained using the labeled pixel dataset. Random Forest was selected for its:

- Robustness to high-dimensional feature spaces
- Ability to handle non-linear relationships between spectral signatures and land cover classes
- Built-in feature importance estimation
- Resistance to overfitting through ensemble averaging
- Computational efficiency for large datasets

The model was configured with the following hyperparameters:

Table 6: Random Forest Hyperparameters

Parameter	Value	Description
n_estimators	100	Number of trees in the forest
max_depth	None	Maximum depth of trees (unlimited)
min_samples_split	2	Minimum samples required to split
min_samples_leaf	1	Minimum samples in leaf node
max_features	sqrt	Features considered per split
bootstrap	True	Bootstrap sampling enabled
random_state	42	Reproducibility seed

2.6 Stage 5: Full Tile Prediction

The trained RF model was applied to all pixels within each of the 10 Sentinel-2 tiles. A validity mask was applied to exclude:

- No-data regions outside the UAE boundary
- Cloud-contaminated pixels not filtered during compositing
- Invalid spectral values (e.g., negative reflectance)

Masked regions were assigned NaN values in the output rasters. This stage produced **10 classified GeoTIFF files**, one per tile.

2.7 Stage 6: Post-Processing

The final stage involved merging the 10 individual tile classifications into a single, seamless UAE-wide classified raster. Post-processing steps included:

1. **Mosaicking:** Combining tile outputs using nearest-neighbor resampling at tile boundaries
2. **No-data handling:** Ensuring consistent no-data values across the mosaic
3. **Coordinate system verification:** Confirming all outputs in WGS84 / UTM projection
4. **Metadata attachment:** Documenting processing parameters and data lineage

The final output represents the spatial distribution of urban, vegetation, desert, and water classes across the entire UAE study area, suitable for subsequent change detection and urban growth analysis.

3 Data Sources and Specifications

3.1 Sentinel-2 MultiSpectral Instrument (MSI)

Table 7: Sentinel-2 MSI Technical Specifications

Attribute	Specification
Satellite	Sentinel-2A / Sentinel-2B
Sensor	MultiSpectral Instrument (MSI)
Spatial Resolution	10 m (B2, B3, B4, B8), 20 m (B11)
Revisit Frequency	5 days (combined constellation)
Spectral Bands	13 bands (Visible to SWIR)
Radiometric Resolution	12-bit
Data Product	Level-2A (Surface Reflectance)
Data Provider	Copernicus Open Access Hub / GEE

3.2 OpenStreetMap (OSM) Data

Table 8: OSM Data Specifications

Attribute	Specification
Data Source	OpenStreetMap (openstreetmap.org)
Data Format	Vector (GeoJSON / Shapefile)
License	Open Database License (ODbL)
Update Frequency	Continuous (community-driven)
Feature Types	Landuse, Natural, Water
Extraction Tool	Overpass API / Geofabrik

3.3 Global Human Settlement Layer (GHSL)

The GHSL dataset provides a global spatial information layer on human presence, derived from satellite imagery and census data. In this workflow, GHSL serves as an auxiliary feature to enhance the detection of built-up areas, particularly in regions where OSM road coverage may be incomplete.

3.4 Data Preprocessing Summary

Table 9: Preprocessing Steps and Parameters

Step	Parameter	Description
Cloud Masking	<20% cloud cover	FMask algorithm on individual scenes
Compositing	Annual median	Reduces cloud contamination and seasonal variation
Resampling	10 m	B11 resampled to 10 m to match other bands
Reprojection	UTM Zone 40N	Consistent coordinate reference system
Tiling	10 tiles	Equal-area division of UAE extent

4 Machine Learning Implementation

4.1 Environment Setup

The machine learning pipeline was implemented in a Kaggle notebook with the following environment configuration:

- **Platform:** Kaggle Notebooks
- **Language:** Python 3.x
- **Key Libraries:** scikit-learn, rasterio, numpy, pandas, geopandas, matplotlib
- **Compute:** CPU-based training (GPU optional for larger datasets)

4.2 Training Data Characteristics

Table 10: Training Dataset Summary

Class	Pixel Count	Percentage	Description
1 – Urban	1,080,777	64.0%	Built-up and
2 – Vegetation	14,689	0.9%	Green spaces and agriculture
3 – Desert	499,543	29.6%	Sand dunes and gravel plains
4 – Water	92,653	5.5%	Water bodies
Total	1,687,662	100.0%	

4.3 Model Training Procedure

The Random Forest classifier was trained using the following procedure:

Algorithm 1 Random Forest Training Algorithm

Require: Labeled pixel dataset $D = \{(x_i, y_i)\}_{i=1}^n$

Require: Number of trees $T = 100$

- 1: Initialize empty forest $F \leftarrow \{\}$
 - 2: **for** $t = 1$ **to** T **do**
 - 3: Bootstrap sample D_t from D with replacement
 - 4: Grow decision tree h_t on D_t using random feature subsets
 - 5: Add h_t to forest: $F \leftarrow F \cup \{h_t\}$
 - 6: **end for**
 - 7: **return** Trained forest F
-

4.4 Feature Importance Analysis

Random Forest provides intrinsic feature importance scores based on the mean decrease in impurity (Gini importance). These scores indicate which spectral bands and indices contribute most to accurate classification.

Table 11: Feature Importance Ranking

Rank	Feature	Importance
1	B11 (SWIR)	0.1810
2	NDVI	0.1707
3	GHSL	0.1504
4	B4 (Red)	0.1163
5	B8 (NIR)	0.1017
6	BSI	0.0731
7	B2 (Blue)	0.0720
8	B3 (Green)	0.0668
9	MNDWI	0.0340
10	NDBI	0.0338

4.5 Model Evaluation

Model performance was assessed using stratified cross-validation and the following metrics:

- **Precision:** Proportion of correctly predicted positive observations
- **Recall (Sensitivity):** Proportion of actual positives correctly identified
- **F1-Score:** Harmonic mean of precision and recall
- **Support:** Number of actual occurrences of each class

4.5.1 Classification Report

Table 12: Classification Report – Precision, Recall, F1-Score, and Support

Class	Precision	Recall	F1-Score	Support
Urban (1)	1.00	0.95	0.97	216156
Vegetation (2)	0.23	0.78	0.36	2938
Desert (3)	1.00	1.00	1.00	99909
Water (4)	0.82	1.00	0.90	18530
Accuracy			0.96	337533
Macro Avg	0.76	0.93	0.81	337533
Weighted Avg	0.98	0.96	0.97	337533

4.5.2 Confusion Matrix

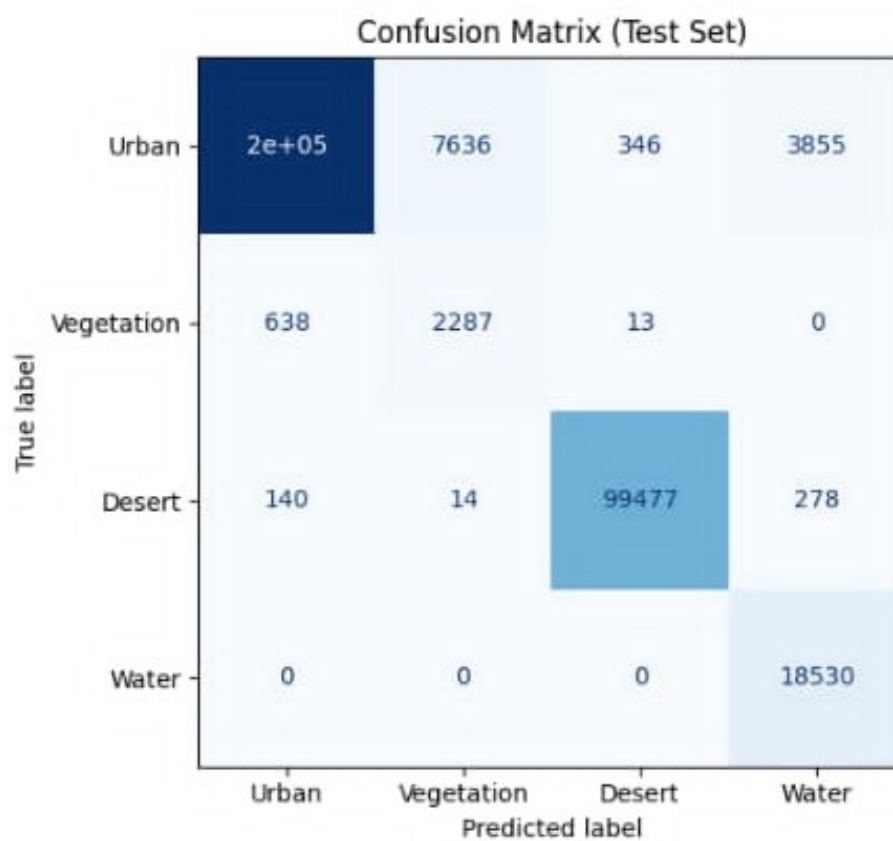


Figure 2: Confusion Matrix for the Random Forest Classifier

5 Results and Analysis

5.1 Classification Outputs

The workflow produced classified land cover maps for 2018, 2021, and 2024. Each map contains four primary classes encoded as follows:

- **Class 1 (Urban):** Represented in red tones
- **Class 2 (Vegetation):** Represented in green tones
- **Class 3 (Desert):** Represented in beige/yellow tones
- **Class 4 (Water):** Represented in blue tones

5.2 Areal Statistics by Year

Table 13: Land Cover Area Comparison: 2018, 2021, and 2024

Land Cover Class	2018 (km ²)	2021 (km ²)	2024 (km ²)	Change (2018–2024)
Urban/Built-up	12659.88	13260.80	14491.08	+1831.20
Vegetation	1020.40	1038.60	1633.26	+612.86
Desert	64513.38	63961.05	61683.01	-2830.37
Water	117.70	50.91	96.11	-21.59
Total	78311.36	78311.36	77903.46	-407.90

Note: Area statistics were calculated using a 10 m spatial resolution Sentinel-2 raster.

Each pixel represents 900 m² (0.0009 km²).

5.3 Area Distribution Diagrams

The following figures illustrate the proportional area distribution for each land cover class across the three study years.

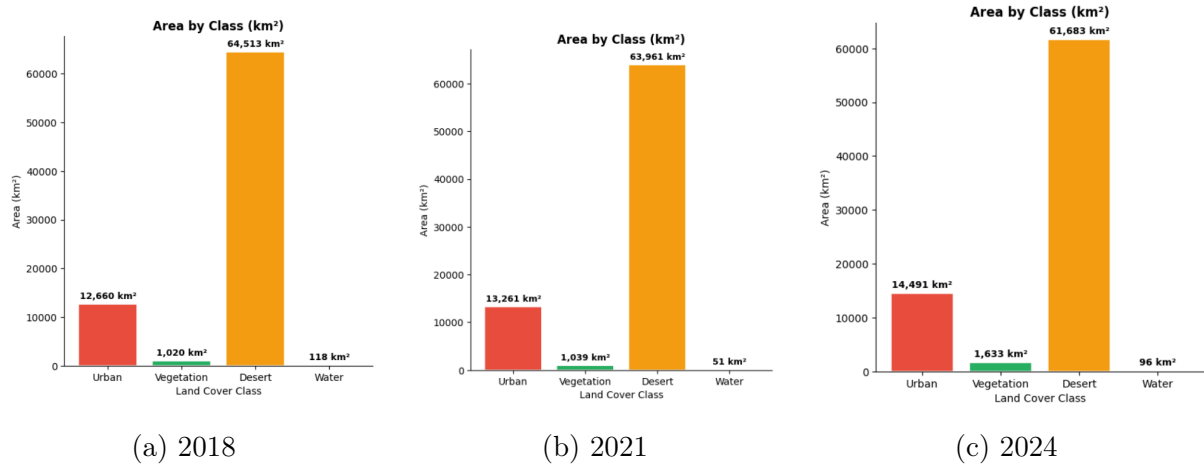


Figure 3: Land Cover Area Distribution by Class for Each Study Year

5.4 Change Detection Analysis

Change detection was performed by comparing classified outputs across three temporal intervals: 2018–2021, 2021–2024, and 2018–2024. This multi-interval approach captures both short-term and long-term land cover dynamics.

5.4.1 2018 to 2021 Change Detection

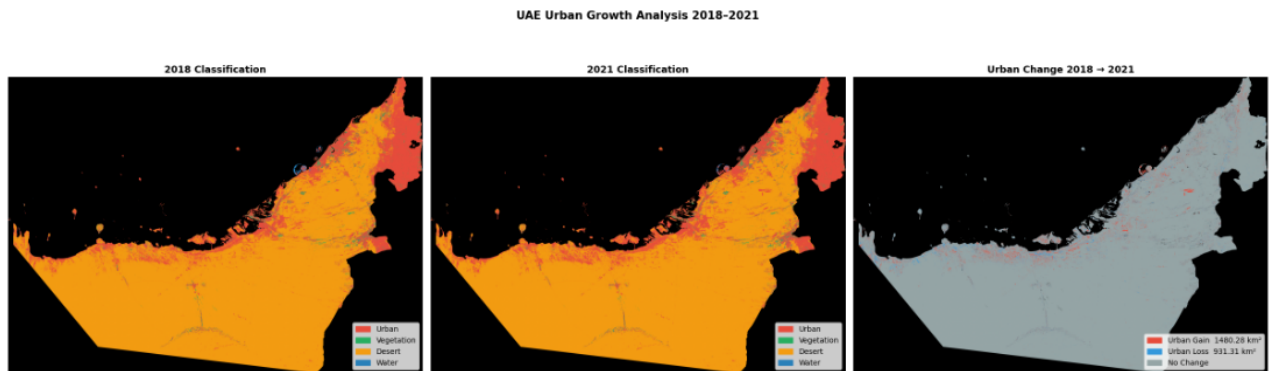


Figure 4: Land Cover Change Detection: 2018 to 2021

The change detection analysis between 2018 and 2021 reveals moderate urban expansion across the UAE. Urban gain reached approximately 1480.28 km^2 , while urban loss accounted for 931.31 km^2 , resulting in a net urban increase of 548.97 km^2 and an overall urban growth rate of 4.7%.

Most newly urbanized areas were concentrated around the metropolitan regions of Dubai, Abu Dhabi, and Sharjah, where infrastructure development and residential expansion were prominent.

sion continued during this period. Coastal development corridors and peri-urban zones exhibited the strongest conversion from desert land to built-up surfaces.

The relatively high amount of urban loss observed in some regions may be attributed to classification inconsistencies, transitional construction zones, or temporary spectral confusion between desert surfaces and urban materials. Nevertheless, the dominant trend remains a clear expansion of urban land cover between 2018 and 2021.

5.4.2 2021 to 2024 Change Detection

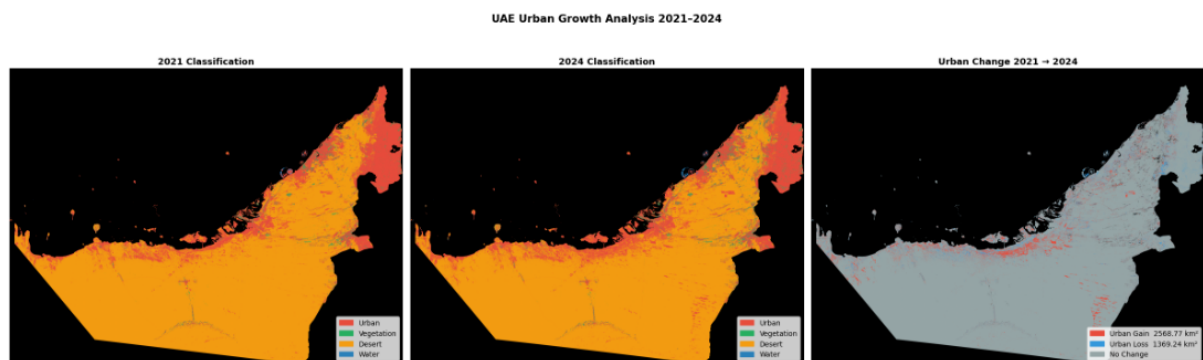


Figure 5: Land Cover Change Detection: 2021 to 2024

Between 2021 and 2024, urban expansion accelerated significantly compared to the previous interval. The analysis identified approximately 2568.77 km^2 of urban gain and 1369.24 km^2 of urban loss, producing a net increase of 1199.52 km^2 and an urban growth rate of 10.0%.

This period corresponds to intensified urban and infrastructure development across several emirates, particularly in Dubai and Abu Dhabi. Large-scale projects, industrial zones, transportation infrastructure, and suburban residential developments contributed to the observed growth patterns.

The results also indicate continued reduction of desert surfaces near major urban corridors, reflecting land conversion for urban use. Vegetation areas showed a slight increase, potentially linked to landscaping projects, irrigated green spaces, and agricultural activities associated with urban development initiatives.

5.4.3 2018 to 2024 Change Detection

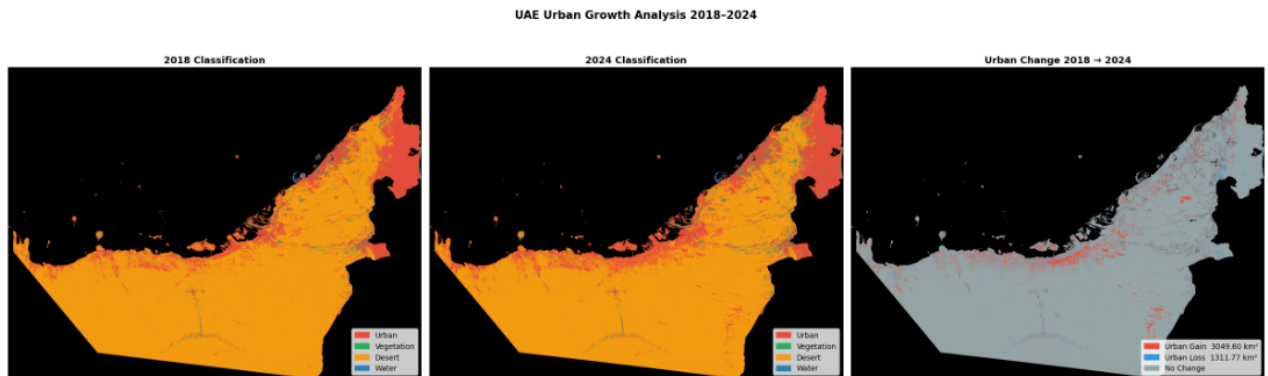


Figure 6: Land Cover Change Detection: 2018 to 2024 (Full Period)

The long-term change detection analysis from 2018 to 2024 demonstrates substantial urban expansion throughout the UAE. Urban gain totaled approximately 3049.60 km², whereas urban loss represented 1311.77 km². This resulted in a net urban increase of 1737.83 km² and an overall urban growth rate of 15.1% during the six-year study period.

The expansion pattern highlights the rapid transformation of desert landscapes into urbanized environments, particularly around existing metropolitan centers and transportation corridors. The strongest urban growth was observed along the Dubai–Abu Dhabi axis, where continuous infrastructure investment and economic development stimulated land conversion.

Desert land remains the dominant land cover class, representing more than 79% of the UAE territory in 2024. However, its gradual reduction confirms the ongoing spatial pressure exerted by urbanization. Vegetation cover also increased between 2018 and 2024, likely reflecting investments in urban greening, parks, and irrigated agricultural zones.

5.5 Urban Growth Analysis

5.5.1 Spatial Patterns of Expansion

The spatial distribution of urban growth reveals that expansion primarily occurred around the UAE's major metropolitan areas, including Dubai, Abu Dhabi, and Sharjah. Urbanization followed major transportation corridors and coastal regions, reflecting the strategic concentration of economic activities and infrastructure development.

Several newly developed zones appeared in peri-urban desert areas, indicating outward urban sprawl and the establishment of new residential, industrial, and commercial districts. Coastal expansion was particularly evident in Dubai and Abu Dhabi, where land reclamation projects and tourism-oriented developments contributed to rapid changes in land cover.

The analysis also suggests densification within existing urban cores, especially in highly developed metropolitan regions where vacant land availability is limited. These patterns align with the UAE's national urban development strategies and large-scale infrastructure investments implemented during the study period.

Key observations to document:

- Directionality of expansion (coastal vs. inland)
- Identification of new development zones
- Comparison with known infrastructure projects
- Density changes in existing urban cores

5.5.2 Growth Rate Calculation

The annual urban growth rate was calculated for each interval:

$$\text{Annual Growth Rate}_{t_1 \rightarrow t_2} = \left(\frac{\text{Area}_{t_2}}{\text{Area}_{t_1}} \right)^{\frac{1}{t_2 - t_1}} - 1 \times 100\% \quad (5)$$

Table 14: Urban Growth Rates by Period

Period	Urban Area (km ²)	Change	Annual Rate
2018	12659.88	—	—
2021	13260.80	+600.92 km ²	1.54%
2024	14491.08	+1230.28 km ²	2.98%

5.6 Vegetation Change Detection

Vegetation cover exhibited a gradual increase throughout the study period, rising from approximately 1020.40 km² in 2018 to 1633.26 km² in 2024. Although vegetation represents a relatively small proportion of the UAE territory due to the country's arid climate, the observed increase is significant in relative terms.

This growth may be associated with urban greening initiatives, development of irrigated parks and recreational areas, landscaping projects, and expansion of agricultural activities. Several metropolitan areas showed localized increases in vegetation cover, particularly around newly developed residential communities and managed green spaces.

Despite this increase, vegetation remains highly fragmented and concentrated in irrigated environments, emphasizing the strong environmental constraints imposed by desert climatic conditions.

Considerations:

- New green spaces and parks
- Agricultural expansion or contraction
- Impact of urbanization on vegetation cover
- Seasonal vs. permanent vegetation changes

5.7 Desert and Water Body Dynamics

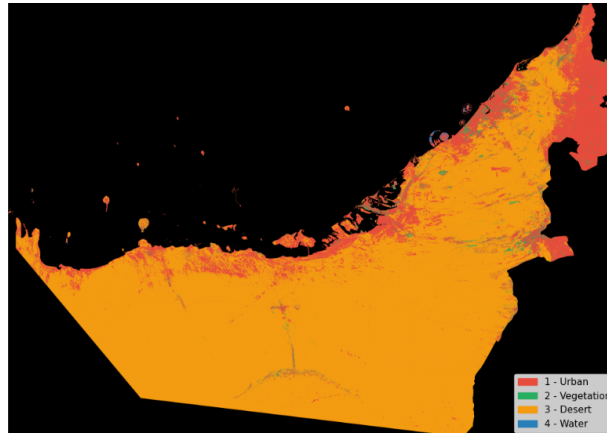
Desert surfaces remained the dominant land cover type throughout the study period, accounting for more than 79% of the UAE territory in all analyzed years. However, the total desert area decreased progressively from 64513.38 km² in 2018 to 61683.01 km² in 2024, primarily due to urban expansion and land conversion processes.

Water bodies represented the smallest land cover class, with relatively minor fluctuations across the study period. Variations in mapped water extent may be related to artificial reservoirs, coastal developments, seasonal effects, or classification uncertainties associated with shallow coastal waters.

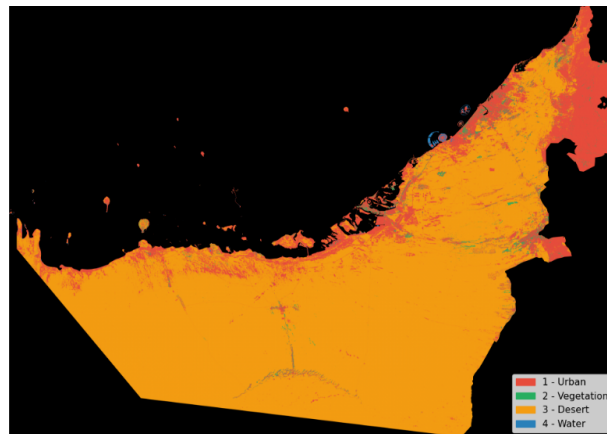
The reduction of desert areas in proximity to major urban centers confirms the strong spatial influence of urban growth on surrounding natural landscapes. These transformations illustrate the rapid anthropogenic modification of arid environments in the UAE.

- Desert encroachment or retreat patterns
- Artificial lake development
- Coastal changes
- Reservoir and dam construction

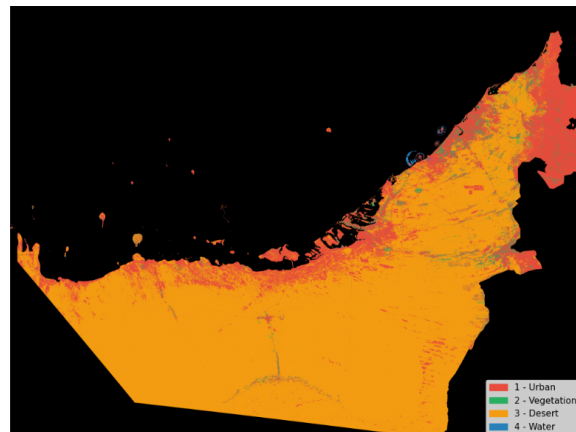
5.8 Visual Comparison of Classifications



(a) 2018 Classification



(b) 2021 Classification



(c) 2024 Classification

Figure 7: UAE Land Cover Classification Maps for 2018, 2021, and 2024

6 Discussion

6.1 Methodological Strengths

1. **Scalability:** The tiling approach enables processing of national-scale datasets on standard computational resources.
2. **Reproducibility:** The use of open-source data (Sentinel-2, OSM) and documented workflows ensures reproducibility.
3. **Cost-effectiveness:** All data sources are freely available, eliminating licensing costs.
4. **Automation:** The pipeline can be re-run for additional years with minimal modification.
5. **Four-Class Scheme:** The inclusion of a dedicated desert class significantly improves the completeness and accuracy of the land cover map in this arid environment.

6.2 Methodological Limitations

1. **OSM Label Quality:** Training label accuracy depends on OSM data completeness and currency. Undersampled or misclassified OSM features propagate errors into the classification.
2. **Spectral Confusion:** In arid environments, built-up areas and bare soil/desert can exhibit similar spectral signatures, particularly in SWIR bands, potentially causing misclassification.
3. **Temporal Resolution:** Annual composites smooth over seasonal variations, which may mask temporary changes (e.g., seasonal vegetation, construction phases).
4. **Spatial Resolution:** The 10 m resolution of Sentinel-2 may not capture fine-scale urban features such as individual buildings.

5. **Class Imbalance:** The vegetation class represents only 0.9% of training labels, which may limit the model's ability to generalize for this class.

6.3 Comparison with Previous Work

The current study builds upon a previous first-semester project developed entirely within Google Earth Engine (GEE). Table 15 summarizes the main differences and improvements between the two workflows.

Table 15: Comparison Between Previous and Current Workflow

Aspect	Previous Work (Semester 1)	Current Work
Processing Environment	Fully implemented in Google Earth Engine	Hybrid workflow using Google Earth Engine and Kaggle
Study Years	Annual analysis from 2018 to 2024	Multi-temporal analysis for 2018, 2021, and 2024
Classification Method	Random Forest with rule-based correction	Enhanced Random Forest with advanced raster processing
Training Samples	2,258 manually digitized samples	Large rasterized training dataset with stratified sampling
Feature Set	Sentinel-2 bands with NDVI, NDBI, MNDWI, BUI, and DUI	Sentinel-2 bands with NDVI, NDBI, MNDWI, BSI, and GHSL
Accuracy	90.8% overall accuracy with $\kappa = 0.859$	96% overall accuracy
Urban Statistics	Some temporal inconsistencies observed, including negative urban growth in 2019	More coherent and realistic urban growth trends between years
Post-Processing	Rule-based correction performed entirely in GEE	Tile mosaicking, masking, and detailed statistical analysis
Visualization	Interactive GEE dashboard visualization	Classification maps, dashboards, and detailed change detection outputs
Main Contribution	Fully cloud-based urban growth monitoring	Improved reliability and detailed urban change analysis

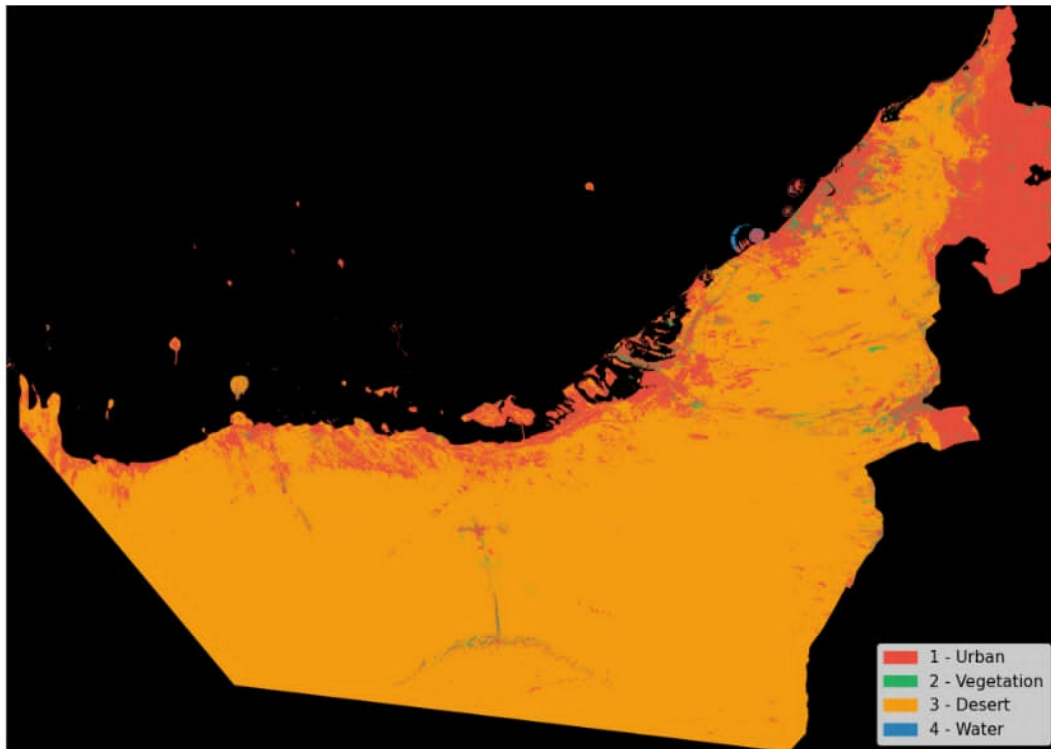


Figure 8: 2019 new classification

The updated workflow produced more stable and realistic urban growth statistics, with urban area increasing from 12,659.88 km² in 2018 to 14,491.08 km² in 2024, confirming sustained urban expansion across the UAE.



Figure 9: Land Cover Change Detection Dashboard: Part 1

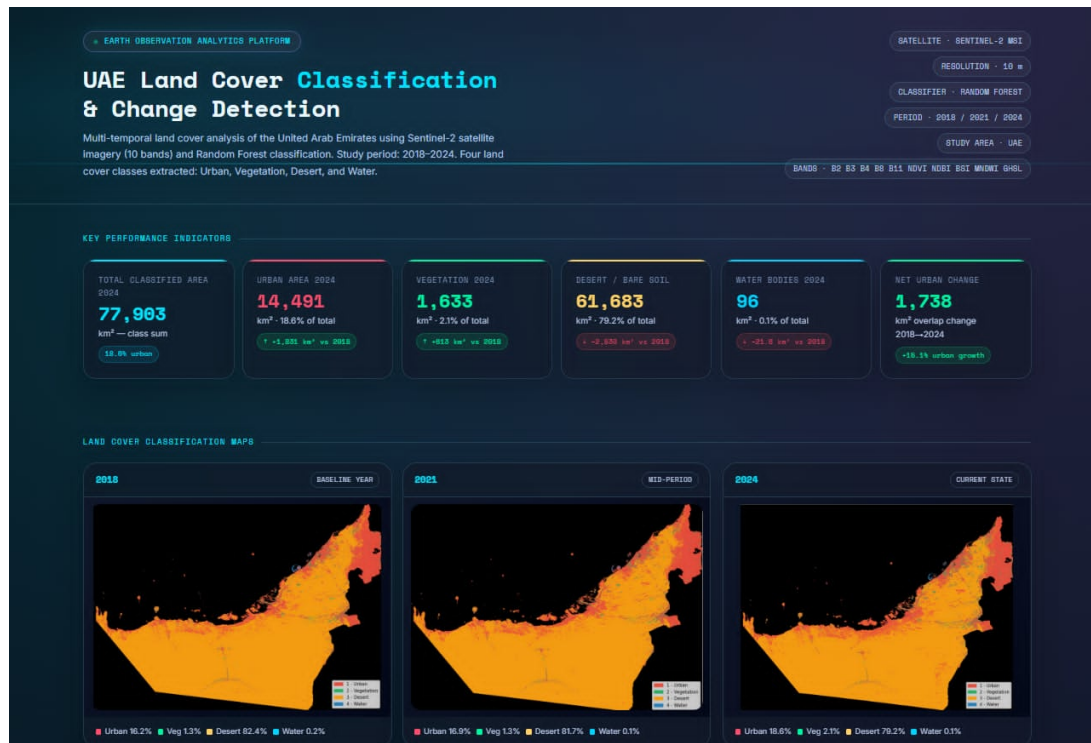


Figure 10: Land Cover Change Detection Dashboard: Part 2

6.4 Implications for Urban Planning

The classification results provide actionable insights for:

- **Infrastructure Planning:** Identifying corridors of urban expansion for proactive service provision
- **Environmental Monitoring:** Tracking vegetation loss and green space preservation
- **Desert Management:** Monitoring desert encroachment and land degradation
- **Water Resource Management:** Monitoring artificial water bodies and coastal dynamics
- **Sustainability Assessment:** Quantifying urban sprawl and its environmental impacts

7 Conclusion and Future Work

7.1 Summary of Findings

This capstone project successfully developed and implemented a comprehensive workflow for land cover classification and urban growth analysis in the United Arab Emirates using multi-temporal Sentinel-2 satellite imagery. The key achievements include:

1. Development of a six-stage processing pipeline integrating GEE, OSM, manual digitization, and machine learning
2. Production of classified land cover maps for 2018, 2021, and 2024 at 10 m spatial resolution with four classes
3. Quantification of urban expansion, vegetation dynamics, desert changes, and water body dynamics across three time periods
4. Multi-interval change detection analysis (2018–2021, 2021–2024, 2018–2024)
5. Demonstration of the feasibility of open-source, cloud-based approaches for national-scale land cover monitoring

7.2 Recommendations for Future Work

1. **Temporal Enhancement:** Extend the analysis to include annual classifications to capture year-to-year growth dynamics.
2. **Accuracy Assessment:** Conduct independent validation using high-resolution reference data (e.g., Google Earth imagery, field GPS points) to quantify classification accuracy.
3. **Deep Learning Exploration:** Compare Random Forest performance with Convolutional Neural Networks (CNNs) or U-Net architectures, which may better capture spatial context.
4. **Sub-class Differentiation:** Distinguish between urban sub-classes (residential, commercial, industrial) and vegetation types (natural vs. managed) for more detailed analysis.

5. **Integration with Socioeconomic Data:** Correlate urban growth patterns with population data, economic indicators, and policy documents to understand drivers of change.
6. **Automation and Deployment:** Package the workflow into a reproducible pipeline for operational monitoring.

7.3 Final Remarks

The methodology presented in this report demonstrates a robust, scalable, and cost-effective approach to monitoring urban growth dynamics in rapidly developing arid regions. The integration of freely available satellite imagery, crowd-sourced geographic data, manually digitized reference polygons, and machine learning techniques represents a paradigm shift in land cover monitoring, enabling researchers and planners to track environmental changes with unprecedented temporal and spatial detail. As the UAE continues its rapid development trajectory, such monitoring systems will be essential for ensuring sustainable urban growth and environmental stewardship.

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